CHAPTER 25 NUCLEAR PHYSICS

Key idea: the mass defect of a nucleus is its binding energy; unstable nuclei decay randomly.

1 THE NUCLEUS

$${}^{A}_{Z}\mathrm{X}$$

- Z: protons; A: nucleons; $N = A - Z$
- Isotopes: same Z, different A
- $R = R_0 A^{1/3}$, $R_0 \approx 1.2$ fm
- Nuclear density the same for all nuclei

2 MASS DEFECT & BINDING ENERGY

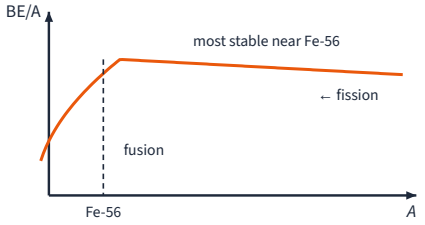
$$\Delta m = Z m_p + (A - Z) m_n - m_{\text{nucleus}}$$

$$B = \Delta m\, c^2$$

$$1\ \text{u} = 931.5\ \text{MeV}$$

Binding energy: energy needed to separate a nucleus into its nucleons.

3 BINDING ENERGY PER NUCLEON



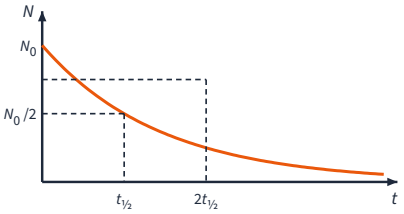
- Peak ≈ 8.8 MeV near ^{56}Fe: most stable
- Light nuclei → **fusion**; heavy nuclei → **fission**

4 RADIOACTIVE DECAY

Emission	Change
α ($^4_2$He)	$A - 4$, $Z - 2$
β^-	A same, $Z + 1$
γ	no change

- **Spontaneous**: not affected by conditions
- **Random**: cannot predict which nucleus decays next

5 DECAY LAW



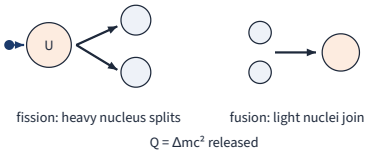
$$N = N_0 e^{-\lambda t} \qquad A = \lambda N$$

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

$$N = N_0 \left(\tfrac{1}{2}\right)^{t/t_{1/2}}$$

$\ln A$ against t: straight line, gradient $-\lambda$. Activity in Bq.

6 NUCLEAR REACTIONS



$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}}\right) c^2$$

Conserved: A, Z (charge), energy, momentum. $Q > 0$: energy released.

7 FISSION VS FUSION

	Fission	Fusion
Process	heavy nucleus splits (e.g. ^{235}U + n)	light nuclei join (e.g. ^{2}H + ^{3}H)
Needs	slow neutron; chain reaction	very high temperature
Energy	≈ 200 MeV per fission	more per kg of fuel
Why	products have higher B/A	products have higher B/A

8 COMMON SECTION A TRAPS

- Using total binding energy instead of **per nucleon** for stability.
- Forgetting electron masses when using atomic masses.
- Taking $t_{1/2} = \lambda \ln 2$ (it is $\ln 2 / \lambda$).
- Mixing u, kg and MeV in one line.
- Time in years with λ in s^{-1}.

EXAM TIPS

- ✓Balance A and Z before finding Q.
- ✓Use 1 u = 931.5 MeV to skip c^2.
- ✓Count half-lives when t is a whole multiple of $t_{1/2}$.
- ✓Answer "why energy is released" with binding energy per nucleon.